3 physical layer : guided media

Problem 3.1 : a signal of power $P_{in} = 10\ mW$ is applied to a 5 km fiber optic fiber. cable has attenuation of 0.3 dB/km

a) total attenuation in dB over 5 km

$$loss = \alpha \cdot d = 5\ km \cdot 0.3\ \frac{dB}{km} = 1.5\ dB$$

b) total power output P_{out} in mW

~~$P_{out} = P_{in} - loss$~~ → loss in mW

$$1.5 = 10 \log_{10}(P)$$

$$0.15 = \log_{10} P$$

$$10^{0.15} = P$$

$$P = 1.41$$

$$P_{out} = 10\ mW - 1.41\ mW$$

$$P_{out} = 8.59$$

→ cannot subtract mW → calculate in dBm & switch

$$P[dBm] = 10 \log_{10}\left(\frac{10\ mW}{1\ mW}\right)$$

$$= 10 \cdot 1 = 10\ dBm$$

$$P_{out}[dBm] = P_{in}[dBm] - loss[dB]$$

$$= 10 - 1.5$$

$$= 8.5\ dBm$$

convert to mW:

$$8.5 = 10 \log_{10}(P)$$

$$P = 10^{0.85}$$

$$P = 7.08$$

c) express the input power in dBm

$$10\ mW \rightarrow dBm$$

$$P[dBm] = 10 \log_{10}\left(\frac{10\ mW}{1\ mW}\right)$$

$$P[dBm] = 10 \log_{10}(10)$$

$$P[dBm] = 10$$

Problem 3.2 : a digital subscriber line (DSL) channel has a bandwidth of 1 MHz an an SNR of 40 dB

a) convert the SNR from dB to a linear ratio

$$SNR_{linear} = 10^{40/10} = 10^4$$

b) compute the shannon channel capacity C

$$C = B \log_2(1 + SNR) = 10^6\ Hz \log_2(1 + 10^4) = 1.33 \times 10^7\ \frac{b}{s}$$

(SNR: linear)

c) a modem achieves 20 Mb/s on this channel, how does this compare to capacity and is it physically possible?

Problem 3.3 :

a) describe the difference between unipolar NRZ and polar NRZ-L. Which is preferred

Unipolar NRZ → 1 → +V, 0 → 0

Polar NRZ → 1 → +V, 0 → -V → preferred because it keeps a balanced DC

b) in manchester encoding, how is a binary '1' represented and '0'

1 → ⎍ → up 0 → ⎍ → down

c) a binary signal uses polar NRZ at 100 Mb/s. what is symbol rate?
if instead 16 PAM is used at the same symbol rate, what is the bit rate?

d) name one advantage / dis of 16 PAM vs binary SNR